Individual Report

MiniOS Alpha | Operating Systems Final Project

Student Name:

Group: Marquez and Friends

Assigned Role: File Management

What did I contribute to the project?

I designed and built the virtual file system in filesystem.py. It lets the user create folders and files, read and edit them, move between directories, rename items and delete them. Everything is kept in memory, so the real disk is never touched.

What features did I personally develop?

- A Node class for files and directories, with a parent link and a dictionary of children
- A FileSystem class with a root directory and a current directory
- dir, mkdir, create, read and delete (the required commands)
- write, rename and cd (extra commands)
- Path building for the prompt and path resolution for cd, including relative paths, absolute paths and ..
- Name checks and protection against duplicate names

What problems did I encounter?

- cd had to handle paths like .. and absolute paths, and not go above the root.
- Users could type names with characters that are not allowed in file names.
- A directory could be deleted while it still held files, or a directory could be read as if it were a file.
- Listing the items in a useful order.

How did I solve those problems?

- I wrote a path resolver that walks one part at a time, handles .. by moving to the parent, and stops at the root.
- I added a name check that rejects empty names, the characters \ / : * ? " < > |, and the names . and ..
- delete refuses non-empty directories, and read, write and cd each check whether the item is a file or a directory first.
- dir sorts folders first, then files by name, and shows the size in bytes.

What did I learn from this project?

I learned that a file system is a tree, and that most of the work is in the rules that keep the tree consistent: unique names, valid paths, and different behaviour for files and directories. I also saw why real operating systems protect directories that still contain files.

Evidence of my contribution:

- filesystem.py (Node and FileSystem classes)
- The file commands in main.py: dir, mkdir, create, read, write, delete, rename, cd

Sample output from my part of the program:

```
MiniOS \> mkdir Documents
Directory created.

MiniOS \> create notes.txt
File created.

MiniOS \> read missing.txt
Error: File does not exist.

MiniOS \> delete Documents
Error: Directory is not empty.
```

Screenshots (attach in the screenshots folder and list the file names here):